

"Ss. Cyril and Methodius" University in Skopje
FACULTY OF COMPUTER
SCIENCE AND ENGINEERING

FACULTY OF COMPUTER SCIENCE AND ENGINEERING

Advanced Programming

Introduction

FCSE 2026

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Prof. Ivan Kitanovski

Assist. Stefan Andonov
Assist. Ana Todorovska

Course Objectives



- To introduce students to the **paradigm of generic programming**. To familiarize them with **abstract data types**, and the creation of **generic classes and functions**. To develop **skills for using data collections** such as **vectors, queues, lists, sets, and dictionaries**. To introduce some aspects of **functional programming** (streams in Java). To apply object-oriented design patterns in modeling and design of a robust program solution.

General course info



- Lectures: 2 hours per week
  - Wednesday 12-14
- Tutorials and exercises:
  - tutorials – 2 hours per week
  - Lab. exercises – 2 hours per week ???



- Self-study

  xyz hours weekly
- Consultations
  - cabinet #219 (MF/FEIT main building)
  - "office" hours: https://consultations.finki.ukim.mk/display/dejan.gjorgjevikj
  - or by appointment (e-mail)



Right to take the exam (signature)

- Activity on the lab exercises
- TBA…

Exams and scoring





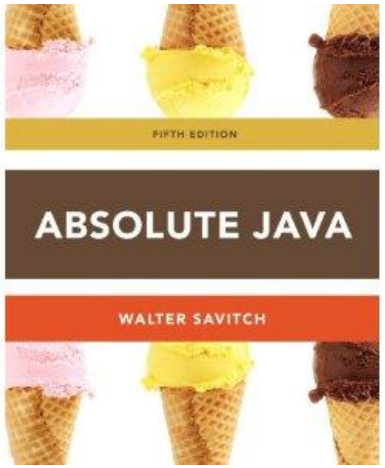
- Two partial exams (45% each) or one final exam (90%)
- Laboratory exercises (10%)
  - Part of task that was on the lab exercises is one of the problems given on the exam (worth 10%)
- Passing minimum 50%
- Bonus points

Course content

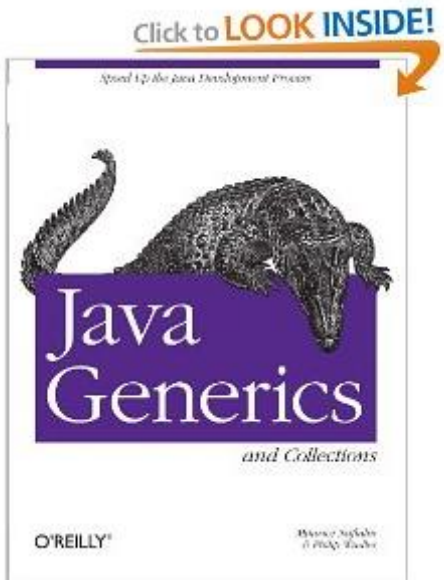
- Classes, interfaces, objects, inheritance, polymorphism, encapsulation
- Handling exceptions, File I/O
- Generics and generic programming
- Data Structures and Their Implementation with Generics
- Collections, collections interfaces, implementation of collections
- Collectible classes
- Data representation
- Refactoring
- Basics of concurrent programming
- Software Design Patterns

Literature (1)

We do not follow any particular book for the lectures on this subject!

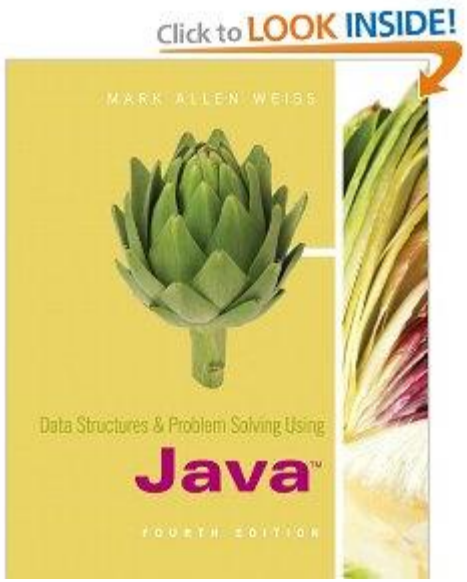


- Walter Savitch and Kenrick Mock., **Absolute JAVA**, Addison-Wesley, 2016.

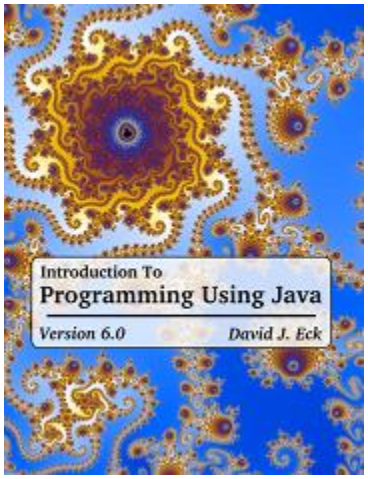


- Naftalin and Philip Wadler, **JAVA Generics and Collections**, O'Reilly, 2006.

Literature (2)

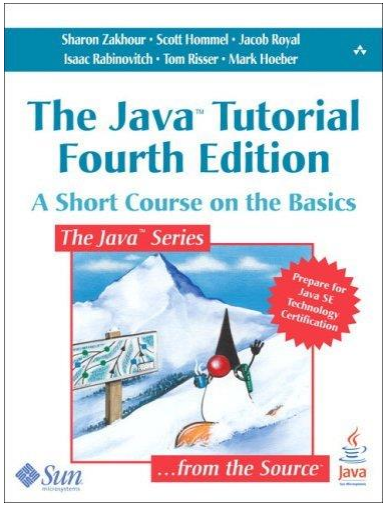


- Mark Allen Weiss, **Data Structures & Problem Solving Using Java**, Addison-Wesley, 2009



- David J. Eck, **Introduction to Programming Using Java**, 6th Edition, 2011. http://www.BruceEckel.com

Literature (3)

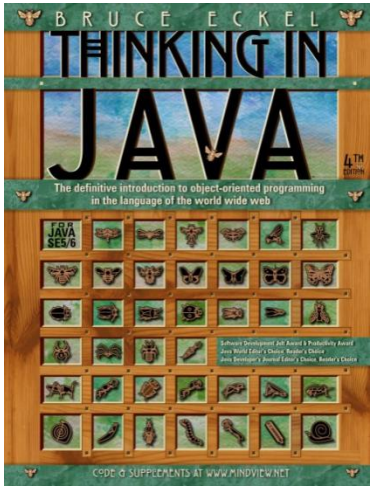


- **The Java Tutorial**, Oracle
  - http://docs.oracle.com/javase/tutorial/



- **Java 8 in Action: Lambdas, Streams, and Functional-style Programming,** Manning, 2015.

Literature (4)

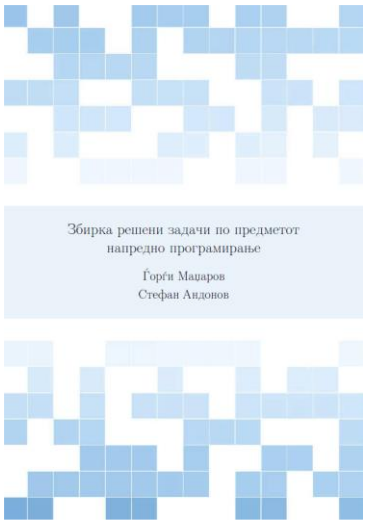


- Thinking in Java, 4th edition
  - by Bruce Eckel
  - http://mindview.net/Books/TIJ4



- Head first design patterns
  - by Eric Freeman, Elisabeth Robson, Bert Bates, Kathy Sierra
  - O'Reilly Media 2004

Literature (5)



- Збирка решени задачи по предметот Напредно програмирање
  - Ѓорѓи Маџаров и Стефан Андонов
  - Јуни, 2021

https://www.ukim.edu.mk/e-izdanija/FINKI/Zbirka_resheni_zadachi_po_napredno_programiranje.pdf

Web resources...

- JAVA tutorials
  - http://docs.oracle.com/javase/tutorial/index.html
- JAVA generics (update)
  - http://docs.oracle.com/javase/tutorial/java/generics/
- JAVA collections
  - http://docs.oracle.com/javase/tutorial/collections/index.html


Questions?